

- 2 (a) (i) reduction in energy (of the oscillations) (B1)
 reduction in amplitude / energy of oscillations (B1)
 due to force (always) opposing motion / resistive forces (B1) [2]
 any two of the above, max 2
- (ii) amplitude is decreasing (very) gradually / oscillations would
 continue (for a long time) / many oscillations M1
 light damping A1 [2]
- (b) (i) frequency = $1 / 0.3$
 = 3.3 Hz A1 [1]
 allow points taken from time axis giving $f = 3.45$ Hz
- (ii) energy = $\frac{1}{2}mv^2$ and $v = \omega a$ C1
 = $\frac{1}{2} \times 0.065 \times (2\pi/0.3)^2 \times (1.5 \times 10^{-2})^2$ M1
 = 3.2 mJ A0 [2]
- (c) amplitude reduces exponentially / does not decrease linearly M1
 so will be not be 0.7 cm A1 [2]

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2 (c) (i) 1 dec C corresponding to $(2840 - 100) / 100 \text{ } ^\circ\text{C}$

C1